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Cyber atack Detection on Nginx Log Data Using XGBoost

Firstyatama Auzan Farhadi[a1], Tsalits Sabila Fatah[a2]

[a]Informatics, Telkom University
Surabaya, Indonesia
[1]firstyatama@student.telkomuniversity.ac.id
[2]tsalitsfatah@ student.telkomuniversity.ac.id

Abstract

This research develops a cyberattack detection model based on Extreme Gradient Boosting (XGBoost) algorithm using NginX log analysis. The dataset of 200 thousand entries is processed through feature engineering techniques, such as the extraction of URL length and suspicious keyword patterns. The evaluation results show that the model accuracy is 94%, with the best performance in the detection of Brute Force and Directory Traversal attacks. However, the model has difficulty in detecting normal activity and SQL Injection, which is due to the imbalance of data distribution. This research provides insight into the importance of dataset balancing and feature enrichment to improve attack detection accuracy in web applications, as well as the potential for real-time implementation of the model to improve cybersecurity.

Keywords: *XGBoost, NginX Logs, Cyber Attack, Security, Confusion matrix*

1. Introduction

In the era of digital transformation, the threat of cyberattacks is increasingly becoming a major concern in various sectors. The Internet, which serves as the main foundation for various digital services and infrastructure, brings with it the risk of cyberattacks such as brute force, directory traversal, and SQL injection. The impact of such attacks not only results in financial losses, but also threatens individual privacy and overall system integrity. As the volume of data continues to grow, the development of efficient solutions to detect these threats has become an urgent need[1], [2].

As the main gateway of communication on the internet, web servers generate data logs that contain a lot of valuable information. If analyzed with the right techniques, these logs can reveal patterns of suspicious activity that often go undetected by traditional security systems. One machine learning algorithm that excels in analyzing complex datasets is Extreme Gradient Boosting (XGBoost). It is renowned for its accuracy and computational efficiency, even on large-scale and imbalanced datasets[1], [3].

This research aims to develop an XGBoost-based cyberattack detection model by utilizing NginX log analysis. The model is designed to automatically detect various types of attacks, including SQL Injection, Brute Force, Directory Traversal, and Denial of Service (DoS). Through the application of feature engineering techniques such as URL length analysis and keyword pattern identification, this research focuses not only on model accuracy, but also on ease of interpretation through the identification of important features. This research is expected to

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provide a more sophisticated and real-time approach to detecting cyberattacks, while supporting the development of more robust mitigation measures. With this approach, this research not only enriches insights in the field of cybersecurity, but also opens up implementation opportunities in various web application systems

2. Reseach Methods

2.1. Datasets Analysis

The dataset used in this research is extracted from NginX logs, comprising 200,000 entries. Each entry records various attributes, such as client IP, timestamp, HTTP request, protocol, response size, referrer, user agent, and additional features derived from these fields. The dataset includes labeled data categorized into four classes: Normal, SQL Injection, Brute Force, and Directory Traversal. The dataset is imbalanced, with the majority of entries belonging to Brute Force and Directory Traversal attacks, while Normal and SQL Injection classes are underrepresented. This imbalance impacts the model's ability to accurately classify minority classes. Feature characteristics such as 'url_length', 'response_size', and 'hour' provide valuable context for detecting unusual access patterns. For instance, longer URLs often indicate potential SQL Injection or Directory Traversal attempts. The class label distribution shows that Brute Force constitutes 50% of the dataset, Directory Traversal 30%, Normal 15%, and SQL Injection 5%.

2.2. Pre-Processing

This research uses the NginX log dataset that records User access patterns including information such as client IP, timestamp, and HTTP requests. The dataset consists of 200,000 entries, with 80% as training data and 20% as test data. The pre-processing steps performed include:

a. Time Conversion: The timestamp column is converted into datetime format, and time features such as hours and days of the week are extracted from the data.
b. Encode Categorical Data: Method and protocol columns are converted to numeric values using LabelEncoder.
c. Feature Engineering: URL length is calculated based on the request field and added as a new feature to the dataset.
d. Empty Data Handling: Rows containing empty values are deleted to keep the data clean.
e. Data Normalization: Numeric features such as response_size and url_length are normalized using StandardScaler[2], [3].

2.3. Attack Detection

This model is designed to detect various types of cyberattacks using feature engineering and classification through the XGBoost algorithm. Each attack type is identified based on specific patterns and characteristics extracted from the NginX log data.

a. SQL Injection
   SQL Injection is detected by identifying suspicious keyword patterns such as UNION, SELECT, or special characters that signify SQL comments, like --, #, or /*. These patterns are typically present in malicious requests attempting to manipulate database queries. Features like url_length and method are instrumental in distinguishing SQL Injection attacks from regular activity, as they capture abnormalities in request structure and behavior.

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b. Directory Traversal
Directory Traversal attacks are characterized by attempts to access unauthorized files or directories, often using patterns like ../ or references to sensitive system files such as /etc/passwd. These patterns are monitored in the request field, and the model leverages features like url_length and protocol to differentiate Directory Traversal attacks from legitimate requests. These features reflect the unusual structure and purpose of such malicious access attempts.

c. Brute Force
Brute Force attacks are detected through the analysis of repetitive access patterns, especially in HTTP methods like POST, which are commonly used in login requests. By analyzing temporal features such as hour and day_of_week, the model captures the timing and frequency of repeated failed login attempts. This helps in identifying Brute Force attacks, which are often characterized by high-intensity access attempts within short time frames.

XGBoost is used to classify access patterns into categories: Normal, SQL Injection, Brute Force. The model parameters are optimized through grid search method to obtain the best performance on the training data. 2

2.4. XGBoost

XGBoost was chosen for its ability to handle unbalanced data through an optimized loss function. The model is built through the following steps:

1. Model Initiation
   XGBoost is trained using a dataset divided into training data (80%) and test data (20%) to ensure a representative distribution.
2. Hyperparameter Optimization
   Parameter: 100 estimators, learning rate 0.1, and max depth 5. Optimized using grid search. Evaluation is done based on accuracy through cross-validation.
3. Model Training
   The processed data is fed into the XGBoost model. During the training process, the performance of the model is evaluated using metrics such as accuracy, precison, recall, and F1 score.
4. Accuracy Calculation
   Accuracy is calculated using the confusion matrix with the following formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Where:

- TP (True Positives) : True predictions for positive classes
- TN (True Negatives): Correct prediction for the negative class
- FP (False Positives): False predictions for the positive class
- FN (False Negatives): False predictions for negative classes

3. Result and Discussion

The model is evaluated using accuracy, precision, recall, F1-score, and confusion matrix metrics. Based on the evaluation results with datasets that have been processed according to the model implementation, the classification report is as follows:

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Table 1. Accuracy Result

Attack	Precison	Recall	F1-Score
Brute Force	0.94	0.99	0.97
Directory Traversal	0.99	0.93	0.96
Normal	0.65	0.16	0.26
SQL Injection	0.82	0.60	0.69
Accuracy			0.94

The model achieved an accuracy of 93.64%. The evaluation results show that the model has a high performance in detecting Brute Force attacks, with a precision of 0.94 and a recall of 0.99. This is supported by the dominant amount of data in this category in the dataset. In the Directory Traversal category, the precision of 0.99 and recall of 0.93 reflect the high detection capability despite the relatively small number of samples. In contrast, in the Normal category, the model only achieved a precision of 0.65 and a recall of 0.16, which indicates the difficulty of the model in recognizing patterns due to the imbalance of data distribution. For the SQL Injection category, the precision of 0.82 and recall of 0.60 show that the model is quite capable of detecting attacks although the limited data distribution causes the results to be less consistent.

The confusion matrix supports these results, where the 'Brute Force' class has a very high recall, while the 'Normal' class shows low performance due to misclassification into other attack categories. The most important features that affect detection include 'hour', 'url_length' and 'response_size'. The 'hour' feature reflects the time of the attack, 'url_length' is an important indicator in detecting SQL Injection and Directory Traversal, while 'response_size' helps distinguish certain attack patterns.



Figure 1. Heatmap Accuracy

Visualization of the confusion matrix in the form of a heatmap shows the distribution of correct and incorrect predictions across classes. This visualization helps identify which classes are frequently misclassified and provides an overview of the model error patterns.

4. Conclusion

This research successfully demonstrated the great potential of using the XGBoost algorithm in cyberattack detection through NginX log analysis. By utilizing feature engineering techniques such as URL length extraction and suspicious keyword pattern analysis, the developed model was able to achieve an excellent accuracy rate of 94%. The best performance is seen in the detection of Brute Force and Directory Traversal attacks, with near perfect precision and recall. This shows that the model is able to identify access patterns that dominate the dataset very effectively.

However, there were significant challenges in detecting normal activity and SQL Injection attacks, as seen by the lower precision and recall for these two categories. This is due to the unbalanced distribution of the data, where the number of normal and SQL Injection data is much less than the other attack categories. This issue underscores the importance of dataset enrichment and the application of balancing techniques such as oversampling or SMOTE to ensure a more even distribution.

The results of this study provide several important insights. First, features such as URL length, access time (hour), and response size (response_size) proved to be significant indicators in identifying specific types of attacks. Secondly, the grid search method used for XGBoost hyperparameter optimization plays a major role in improving the overall performance of the model.

This research opens up further implementation opportunities, especially in developing an attack detection system that can be operated in real-time. With enhancements such as dataset balancing, additional data collection, and integration with existing security systems, this model can become a more robust solution for improving web application security. As a next step, future research can explore other machine learning algorithms such as LSTM or ensemble combinations to overcome the existing weaknesses.

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